TABLE 18-1 MATERIAL PROPERTIES FOR CIRCULAR PLATES

Notation

ν_r, ν_ϕ = Poisson's ratio in r and ϕ directions; for isotropic materials, $\nu_r = \nu_\phi = \nu$
E_r, E_ϕ = modulus of elasticity in r and ϕ directions; for isotropic materials, $E_r = E_\phi = E$
$D_r, D_\phi, D_{r\phi}$ = flexural rigidities
K_r, K_ϕ = extensional rigidities in r and ϕ directions
α_r, α_ϕ = thermal expansion coefficients in r and ϕ directions
h = thickness of plate
G = shear modulus of elasticity, same value for isotropic and orthotropic materials

Plate	Constants
1. Homogeneous isotopic	$\nu_r = \nu_\phi = \nu$ $D_r = D_\phi = D = Eh^3/[12(1-\nu^2)]$ $G = E/[2(1+\nu)]$ $D_{r\phi} = \frac{1}{2}D(1-\nu) = \frac{1}{12}Gh^3$ $K_r = K_\phi = K = Eh/(1-\nu^2)$ $\alpha_r = \alpha_\phi = \alpha$
2. Homogeneous orthotropic	$D_j = E_j h^3/[12(1-\nu_r\nu_\phi)] \qquad j = r, \phi$ $K_j = E_j h/(1-\nu_r\nu_\phi)$ $D_{r\phi} = \frac{1}{12}Gh^3$
3. Continuously composite isotropic	$D_r = D_\phi = D = \int_{-h/2}^{h/2} \frac{Ez^2}{1-\nu^2}\,dz$ $D_{r\phi} = \int_{-h/2}^{h/2} \frac{Ez^2}{2(1+\nu)}\,dz$ $K_r = K_\phi = K = \int_{-h/2}^{h/2} \frac{E}{1-\nu^2}\,dz$ $\alpha_r = \alpha_\phi = \alpha$
4. Continuously composite, orthotropic	$D_j = \frac{1}{1-\nu_r\nu_\phi}\int_{-h/2}^{h/2} \frac{E_j z^2}{1-\nu_r\nu_\phi}\,dz \qquad j = r, \phi$ $K_j = \int_{-h/2}^{h/2} \frac{E_j}{1-\nu_r\nu_\phi}\,dz$ $D_{r\phi} = \int_{-h/2}^{h/2} Gz^2\,dz$

TABLE 18-1 (continued) MATERIAL PROPERTIES FOR CIRCULAR PLATES

Plate	Constants
5. Layered	$D_j = 2\sum_i \left(\frac{E_j}{1-\nu_r\nu_\phi}\right)_i \int_{\Delta h_i} z^2\,dz \qquad j = r, \phi$ $K_j = 2\sum_i \left(\frac{E_j}{1-\nu_r\nu_\phi}\right)_i \Delta h_i$ $D_{r\phi} = 2\sum_i (G)_i \int_{\Delta h_i} z^2\,dz$ Δh_i = thickness of ith layer The summation extends over half of the plate thickness.

TABLE 18-2 STRESSES OF CIRCULAR PLATES

Notation

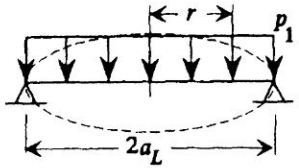
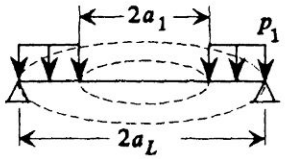
M_r, M_ϕ = Bending moments per unit length
$M_{r\phi}$ = twisting moment per unit length
P_r, P_ϕ = in-plane forces per unit length
T = temperature change
σ_r = radial normal stress (F/L^2)
σ_ϕ = circumferential normal stress (F/L^2)
$\tau_{r\phi}$ = shear stresses(F/L^2)
ν_r, ν_ϕ = Poisson's ratio in r and ϕ directions; for isotropic materials, $\nu_r = \nu_\phi = \nu$
E_r, E_ϕ = modulus of elasticity in r and ϕ directions; for isotropic materials, $E_r = E_\phi = E$
D_r, D_ϕ, $D_{r\phi}$ = flexural rigidities
K_r, K_ϕ = extensional rigidities in r and ϕ directions
α_r, α_ϕ = thermal expansion coefficients in r and ϕ directions
h = thickness of the plate
G = shear modulus of elasticity, $G_{r\phi} = \frac{1}{12}h^2G$

Plate	Stresses
1. Homogeneous isotropic material	$\sigma_r = \dfrac{M_r z}{h^3/12}, \qquad \sigma_\phi = \dfrac{M_\phi z}{h^3/12}, \qquad \tau_{r\phi} = -\dfrac{z M_{r\phi}(1-\nu^2)}{h^3/12}$
2. Other materials; defined in cases 2 and 3 of Table 18-1	$\sigma_r = \bar{E}_r\left[-\dfrac{\bar{P}}{K_r} + \dfrac{\overline{M}_r z}{D_r} - (\alpha_r + \nu_\phi\alpha_\phi)T\right]$ $\sigma_\phi = \bar{E}_\phi\left[-\dfrac{\bar{P}_\phi}{K_\phi} + \dfrac{\bar{M}_\phi z}{D_\phi} - (\alpha_\phi + \nu_r\alpha_r)T\right]$ $\tau_{r\phi} = -G_{r\phi}\dfrac{M_{r\phi}z}{D_{r\phi}}$ where $\bar{P} = P_r + P_T \quad \bar{P}_\phi = P_\phi + P_{T\phi}$ $\bar{M}_r = M_r + M_{Tr} \quad \bar{M}_\phi = M_\phi + M_{T\phi}$ $P_T = \displaystyle\int_{-h/2}^{h/2} \frac{E_r(\alpha_r + \nu_\phi\alpha_\phi)}{1-\nu_r\nu_\phi} T\,dz, \quad M_{Tr} = \int_{-h/2}^{h/2} \frac{E_r(\alpha_r + \nu_\phi\alpha_\phi)}{1-\nu_r\nu_\phi} T\,z\,dz$ $P_{T\phi} = \displaystyle\int_{-h/2}^{h/2} \frac{E_\phi(\alpha_\phi + \nu_r\alpha_r)}{1-\nu_r\nu_\phi} T\,dz, \quad M_{T\phi} = \int_{-h/2}^{h/2} \frac{E_\phi(\alpha_\phi + \nu_r\alpha_r)}{1-\nu_r\nu_\phi} T\,z\,dz$ $\bar{E}_r = \dfrac{E_r}{1-\nu_r\nu_\phi} \qquad \bar{E}_\phi = \dfrac{E_\phi}{1-\nu_r\nu_\phi}$ and $\bar{P}$ and $\bar{P}_\phi$ are the in-plane compression forces. If the forces are tensile, replace $\bar{P}$ and $\bar{P}_\phi$ by $-\bar{P}$ and $-\bar{P}_\phi$.

TABLE 18-3 DEFLECTIONS AND INTERNAL FORCES FOR CIRCULAR PLATES WITH AXIALLY SYMMETRIC LOADS AND BOUNDARY CONDITIONS

Notation

w = deflection
M_r, M_ϕ = bending moments per unit length
r = radial coordinate
a_L = radius of outer boundary
$\alpha = r/a_L$
$D = Eh^3/[12(1-\nu^2)]$
Q_r = transverse shear force per unit length
ν = Poisson's ratio
a_1 = radial location of loading
$\beta = a_1/a_L$
p, p_1 = distributed loading (F/L^2)

Structural System and Static Loading	Deflection and Internal Forces
1. 	$w = \dfrac{p_1 a_L^4}{64D}\left(\dfrac{5+\nu}{1+\nu} - \dfrac{6+2\nu}{1+\nu}\alpha^2 + \alpha^4\right)$ $M_r = \dfrac{1}{16}p_1 a_L^2(3+\nu)(1-\alpha^2)$ $M_\phi = \dfrac{1}{16}p_1 a_L^2[3+\nu-(1+3\nu)\alpha^2]$ $Q_r = -\dfrac{1}{2}p_1 a_L \alpha$
2. 	If $\alpha \le \beta$, $w = \dfrac{p_1 a_L^4}{64D(1+\nu)}(C_1 - 2C_2\alpha^2)$ $M_r = M_\phi = \dfrac{1}{16}p_1 a_L^2 C_2, \quad Q_r = 0$ where $C_1 = 5+\nu-4(3+\nu)\beta^2+(7+3\nu)\beta^4-4(1+\nu)\beta^4\ln\beta$

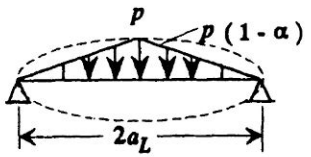
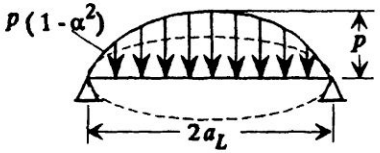
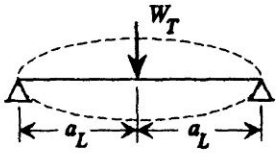
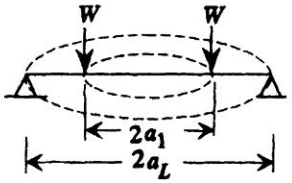
	$C_2 = 3 + \nu - 4\beta^2 + (1-\nu)\beta^4 + 4(1+\nu)\beta^2 \ln\beta$ If $\alpha \geq \beta$, $w = \frac{p_1 a_L^4}{32D(1+\nu)}[(3+\nu)(1-2\beta^2) - (1-\nu)\beta^4] \times (1-\alpha^2)$ $\quad - \frac{p_1 a_L^4}{64D}(1 - \alpha^4 + 8\alpha^2\beta^2 \ln\alpha + 4\beta^4 \ln\alpha)$ $M_r = \frac{1}{16} p_1 a_L^2 \left[\left(3 + \nu - \frac{1-\nu}{\alpha^2}\beta^4 \right)(1-\alpha^2) + 4(1+\nu)\beta^2 \ln\alpha \right]$ $M_\phi = \frac{1}{16} p_1 a_L^2 \left[\left(1 + 3\nu + \frac{1-\nu}{\alpha^2}\beta^4 \right)(1-\alpha^2) + 4(1+\nu)\beta^2 \ln\alpha + 2(1-\nu)(1-\beta^2)^2 \right]$ $Q_r = -\frac{1}{2} p_1 a_1 \left(\alpha - \frac{\beta^2}{\alpha} \right)$
3.  p = maximum value of distributed load	$w = \frac{7pa_L^4}{240(1+\nu)D}(1-\alpha^2) + \frac{pa_L^4}{14{,}400D}(129 - 290\alpha^2 + 225\alpha^4 - 64\alpha^5)$ $(M_r)_{\alpha=0} = (M_\phi)_{\alpha=0} = \frac{1}{720} pa_L^2 (71 + 29\nu)$ $(Q_r)_{\alpha=1} = -\frac{1}{6} pa_L$
4.  	$w = \frac{pa_L^4}{24(1+\nu)D}(1-\alpha^2) + \frac{pa_L^4}{576D}(7 - 15\alpha^2 + 9\alpha^4 - \alpha^6)$ $M_r = \frac{1}{96} pa_L^2 [13 + 5\nu - 6(3+\nu)\alpha^2 + (5+\nu)\alpha^4]$ $M_\phi = \frac{1}{96} pa_L^2 [13 + 5\nu - 6(1+3\nu)\alpha^2 + (1+5\nu)\alpha^4]$ $Q_r = -\frac{1}{6} pa_L \alpha (2 - \alpha^2)$

TABLE 18-3 (continued) DEFLECTIONS AND INTERNAL FORCES FOR CIRCULAR PLATES WITH AXIALLY SYMMETRIC LOADS AND BOUNDARY CONDITIONS

Structural System and Static Loading	Deflection and Internal Forces
5. Concentrated force: Units of total load W_T: force 	$w = \frac{W_T a_L^2}{16\pi D}\frac{(3+\nu)}{(1+\nu)}(1-\alpha^2) + \frac{W_T \alpha_L^2}{8\pi D}\alpha^2 \ln\alpha$ $M_r = -\frac{W_T}{4\pi}(1+\nu)\ln\alpha$ $M_\phi = M_r + \frac{W_T}{4\pi}(1-\nu)$ $Q_r = -\frac{W_T}{2\pi a_L \alpha}$
6. Concentrated force applied on circle: Units of W: force/length 	If $\alpha \le \beta$, $w = \frac{W a_L^2 a_1}{8D}(C_1 - C_2\alpha^2)$ $M_r = M_\phi = \frac{1}{4}W a_1(1+\nu)C_2$ where $C_1 = \frac{3+\nu}{1+\nu}(1-\beta^2) + 2\beta^2 \ln\beta$ $C_2 = \frac{1-\nu}{1+\nu}(1-\beta^2) + 2\ln\beta$ $Q_r = 0$ If $\alpha \ge \beta$, $w = \frac{W a_L^2 a_1}{8D(1+\nu)}(C_3 - C_3\alpha^2 + 2\beta^2 \ln\alpha + 2\alpha^2 \ln\alpha)$

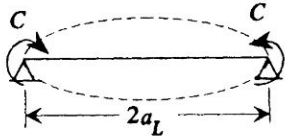
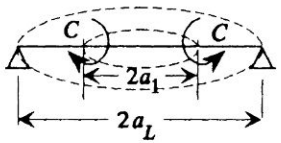
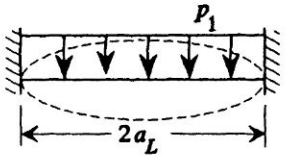
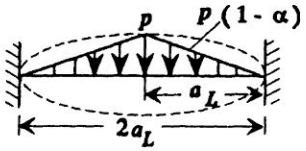
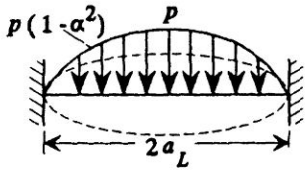
	$C_3 = \dfrac{3+\nu}{1+\nu} - \dfrac{1-\nu}{1+\nu}\beta^2$ $M_r = \dfrac{1}{4}\dfrac{Wa_1}{\alpha^2}[(1-\nu)(1-\alpha^2)\beta^2 - 2(1+\nu)\alpha^2 \ln\alpha]$ $M_\phi = \dfrac{1}{4}\dfrac{Wa_1}{\alpha^2}[2(1-\nu)\alpha^2 - (1-\nu)(1+\alpha^2)\beta^2 - 2(1+\nu)\alpha^2 \ln\alpha]$ $Q_r = -\dfrac{\beta}{\alpha}W$
7. Concentrated moment applied on circle: Units of C: force × length/length  	$w = \dfrac{Ca_L^2}{2D(1+\nu)}(1-\alpha^2)$ $M_r = M_\phi = C$ $Q_r = 0$
8.  	If $\alpha \le \beta$, $w = \dfrac{Ca_L^2}{4D}\left[2\left(\dfrac{1}{1+\nu} - \ln\beta\right)\beta^2 - \left(1 + \dfrac{1-\nu}{1+\nu}\beta^2\right)\alpha^2\right]$ $M_r = M_\phi = \dfrac{C}{2}[1+\nu+(1-\nu)\beta^2]$ $Q_r = 0$ If $\alpha \ge \beta$, $w = \dfrac{Ca_L^2}{D}\beta^2\left[\dfrac{1-\nu}{1+\nu}(1-\alpha^2) - 2\ln\alpha\right]$

TABLE 18-3 (continued) DEFLECTIONS AND INTERNAL FORCES FOR CIRCULAR PLATES WITH AXIALLY SYMMETRIC LOADS AND BOUNDARY CONDITIONS

Structural System and Static Loading	Deflection and Internal Forces
	$M_r = -M_\phi = \frac{C}{2}(1-\nu)(1-\alpha^2)\frac{\beta^2}{\alpha^2}$ $Q_r = 0$
9. 	$w = \frac{p_1 a_L^4}{64D}(1-\alpha^2)^2$ $M_r = \frac{1}{16}p_1 a_L^2[1+\nu-(3+\nu)\alpha^2]$ $M_\phi = \frac{1}{16}p_1 a_L^2[1+\nu-(1+3\nu)\alpha^2]$ $Q_r = -\frac{1}{2}p_1 a_L \alpha$
10. 	$w = \frac{p a_L^4}{14{,}400D}(129-290\alpha^2+225\alpha^4-64\alpha^5)$ $(M_r)_{\alpha=0} = (M_\phi)_{\alpha=0} = \frac{29 p a_L^2}{720}(1+\nu)$ $(Q_r)_{\alpha=1} = -\frac{1}{6}p a_L \qquad (M_r)_{\alpha=1} = (M_\phi)_{\alpha=1} = -\frac{7 p a_L^2}{120}$
11. 	$w = \frac{p a_L^4}{576D}(7-15\alpha^2+9\alpha^4-\alpha^6)$ $M_r = \frac{1}{96}p a_L^2[5(1+\nu)-6(3+\nu)\alpha^2+(5+\nu)\alpha^4]$

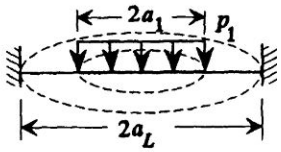
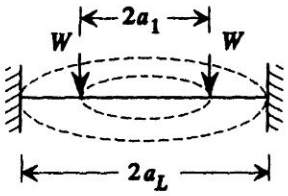
	$M_\phi = \frac{1}{96} p a_L^2 [5(1+\nu) - 6(1+3\nu)\alpha^2 + (1+5\nu)\alpha^4]$ $Q_r = -\frac{1}{4} p a_L (2\alpha - \alpha^3)$
12. 	If $\alpha \le \beta$, $w = \frac{p_1 a_L^4}{64D}[C_1 + 2C_2(1-\alpha^2) + \alpha^4]$ where $C_1 = 4\beta^2 - 5\beta^4 + 8\beta^2 \ln\beta + 4\beta^4 \ln\beta$ $C_2 = \beta^2(\beta^2 - 4\ln\beta)$ $M_r = \frac{p_1 a_L^2}{16}[(1+\nu)(\beta^4 - 4\beta^2 \ln\beta) - (3+\nu)\alpha^2]$ $M_\phi = \frac{p_1 a_L^2}{16}[(1+\nu)(\beta^4 - 4\beta^2 \ln\beta) - (1+3\nu)\alpha^2]$ $Q_r = -\frac{p_1 a_1}{2}\frac{\alpha}{\beta}$ If $\alpha \ge \beta$, $w = \frac{p_1 a_L^2 a_1^2}{32D}[2 - 2\alpha^2 + \beta^2(1 - \alpha^2 + 2\ln\alpha) + 4\alpha^2 \ln\alpha]$ $(M_r)_{\alpha=1} = -\frac{1}{8} p_1 a_1^2 (2 - \beta^2)$ $(M_\phi)_{\alpha=1} = \nu (M_r)_{\alpha=1}$ $Q_r = -\frac{p_1 a_1}{2}\frac{\beta}{\alpha}$

TABLE 18-3 (continued) DEFLECTIONS AND INTERNAL FORCES FOR CIRCULAR PLATES WITH AXIALLY SYMMETRIC LOADS AND BOUNDARY CONDITIONS

Structural System and Static Loading	Deflection and Internal Forces
13. 	If $\alpha \le \beta$, $w = \dfrac{Wa_L^2 a_1}{8D}(C_1 - C_2\alpha^2)$ $M_r = M_\phi = \frac{1}{4}Wa_1(1+\nu)C_2$ $Q_r = 0$ where $C_1 = (1-\beta^2) + 2\beta^2 \ln\beta$ $C_2 = (\beta^2 - 1) - 2\ln\beta$ If $\alpha \ge \beta$, $w = \dfrac{Wa_L^2 a_1}{8D}\left[(1+\beta^2)(1-\alpha^2) + 2\beta\ln\alpha + 2\alpha^2\ln\alpha\right]$ $(M_r)_{\alpha=1} = -\frac{1}{2}W(1-\beta^2) \qquad (M_\phi)_{\alpha=1} = \nu(M_r)_{\alpha=1}$ $Q_r = -\dfrac{W_T\beta}{\alpha}$

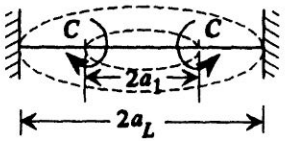
14. 	If $\alpha \le \beta$, $w = -\dfrac{Ca_L^2}{4D}\left[2\beta^2 \ln \beta + (1-\beta^2)\alpha^2\right]$ $M_r = M_\phi = \dfrac{C}{2}(1+\nu)(1-\beta^2)$ If $\alpha \ge \beta$, $w = -\dfrac{Ca_L^2}{4D}\beta^2(1-\alpha^2+2\ln\alpha)$ $(M_r)_{\alpha=1} = -C\beta^2 \qquad (M_\phi)_{\alpha=1} = -\nu C\beta^2$

TABLE 18-4 PART A: CIRCULAR PLATES WITH ARBITRARY LOADING: GENERAL RESPONSE EXPRESSIONS

Notation

a_0 = radius of center hole
w = deflection
θ = slope
V = shear force, $= V_r$
M = bending moment, $= M_r$
w_0, θ_0, V_0, M_0 = initial parameters for w, θ, V, and M
F_w, F_θ, F_V, F_M = loading functions defined in part B
p, p_1, p_2 = distributed applied loading in force/length^2
r = radial coordinate
R = reaction force at center support defined in part C
C_1 = integration constant defined in part C
ν = Poisson's ratio
E = modulus of elasticity
h = thickness of plate
$D = Eh^3/[12(1-\nu^2)]$

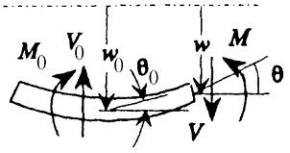
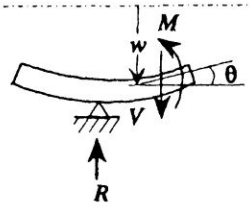
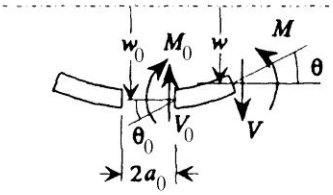
Plate[a]	Response Expressions
1. Plate without center hole 	$w = w_0 - M_0\dfrac{r^2}{2D(1+\nu)} + F_w \qquad \theta = M_0\dfrac{r}{D(1+\nu)} + F_\theta$ $V = F_V \qquad M = M_0 + F_M$
2. Plate with rigid support at center 	$w = -\dfrac{R}{8\pi D}r^2(\ln r - 1) + C_1\dfrac{r^2}{4} + F_w$ $\theta = \dfrac{R}{4\pi D}r\left(\ln r - \dfrac{1}{2}\right) - C_1\dfrac{r}{2} + F_\theta$ $V = \dfrac{R}{2\pi r} + F_V$ $M = \dfrac{R}{4\pi}\left[(1+\nu)\ln r + \dfrac{1-\nu}{2}\right] - \dfrac{C_1 D}{2}(1+\nu) + F_M$

TABLE 18-4 (continued) PART A: CIRCULAR PLATES WITH ARBITRARY LOADING: GENERAL RESPONSE EXPRESSIONS

Plate	Response Expressions
3. Plate with center hole 	$w = w_0 + \theta_0\left[-\frac{1}{2}(1+\nu)a_0 \ln\frac{r}{a_0} - (1-\nu)\frac{r^2-a_0^2}{4a_0}\right]$ $+ M_0\left(\frac{a_0^2}{2D}\ln\frac{r}{a_0} - \frac{r^2-a_0^2}{4D}\right)$ $+ V_0\frac{a_0}{4D}\left[-(a_0^2+r^2)\ln\frac{r}{a_0} + (r^2-a_0^2)\right] + F_w$ $= w_0U_{ww} + \theta_0U_{w\theta} + M_0U_{wM} + V_0U_{wV} + F_w$ $\theta = \theta_0\left[(1+\nu)\frac{a_0}{2r} + (1+\nu)\frac{r}{2a_0}\right]$ $+ M_0\frac{1}{2Dr}(r^2-a_0^2)$ $+ V_0\left[\frac{a_0 r}{2D}\ln\frac{r}{a_0} - \frac{a_0}{4Dr}(r^2-a_0^2)\right] + F_\theta$ $= \theta_0U_{\theta\theta} + M_0U_{\theta M} + V_0U_{\theta V} + F_\theta$ Note that $U_{\theta w} = 0$. $V = V_0\frac{a_0}{r} + F_V = V_0U_{VV} + F_V$ Note that $U_{Vw} = U_{V\theta} = U_{VM} = 0$. $M = \theta_0(1-\nu^2)\frac{Da_0}{2}\left(\frac{1}{a_0^2} - \frac{1}{r^2}\right) + M_0\left(\frac{1-\nu}{2}\frac{a_0^2}{r^2} + \frac{1+\nu}{2}\right)$ $+ V_0\frac{a_0}{2}\left[(1+\nu)\ln\frac{r}{a_0} + (1-\nu)\frac{r^2-a_0^2}{2r^2}\right] + F_M$ $= \theta_0U_{M\theta} + V_0U_{MV} + M_0U_{MM} + F_M$ Note that $U_{Mw} = 0$. U_{ij} are transfer matrix elements.

[a] Responses use the sign conventions in the figures. (Sign convention 1 of Appendix II).

TABLE 18-13 DISPLACEMENTS AND STRESSES FOR LARGE DEFLECTION OF UNIFORMLY LOADED CIRCULAR PLATES

Notation

E = modulus of elasticity
ν = Poisson's ratio
h = thickness of plate
a = radius of plate
w_0 = maximum deflection at center
p_1 = uniformly distributed load on plate

The stresses in this table are at the lower surface. To use this table, solve the relationship between load and deflection for w_0 and then calculate the other responses.

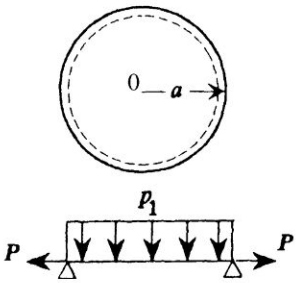
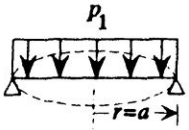
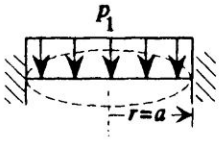
Plates	Deflection and Stresses
1. Simply supported outer edge with in-plane force [18.11]  The radial displacement at the outer edge is zero.	Relationship between load and deflection: $\frac{p_1 a^4}{Eh^4} = \frac{5.333}{1-\nu^2}\frac{w_0}{h} + \frac{2(23-9\nu)}{21(1-\nu)}\left(\frac{w_0}{h}\right)^3$ Extensional radial stress at center: $(\sigma_{re})_0 = \frac{Eh^2}{a^2}\frac{5-3\nu}{6(1-\nu)}\left(\frac{w_0}{h}\right)^2$ Maximum stress in bending at center: $(\sigma_{rb})_0 = \frac{Eh^2}{a^2}\frac{2}{1-\nu}\left(\frac{w_0}{h}\right)$ Maximum total radial stress at center: $\sigma_{r0} = (\sigma_{re})_0 + (\sigma_{rb})_0$ Maximum total radial stress at outer boundary: $\sigma_{ra} = \frac{Eh^2}{a^2}\left[\frac{1}{3(1-\nu)}\left(\frac{w_0}{h}\right)^2 + \frac{4}{1-\nu^2}\frac{w_0}{h}\right]$ Tensile in plane radial outer boundary reaction per unit length: $P = \frac{1}{3(1-\nu)}\frac{Eh^3}{a^2}\left(\frac{w_0}{h}\right)^2$

TABLE 18-13 (continued) DISPLACEMENTS AND STRESSES FOR LARGE DEFLECTION OF UNIFORMLY LOADED CIRCULAR PLATES

Plates	Deflection and Stresses
2. Simply supported outer edge without in-plane force [18.14]  The radial displacement at the outer edge is not restrained. Hence the radial force P is zero.	Relationship between load and deflection: $\frac{p_1 a^4}{Eh^4} = \frac{1.016}{1-\nu}\frac{w_0}{h} + 0.376\left(\frac{w_0}{h}\right)^3$ Maximum radial stress at center: $\sigma_{r0} = \frac{Eh^2}{a^2}\left[\frac{1.238}{1-\nu^2}\frac{w_0}{h} + 0.294\left(\frac{w_0}{h}\right)^2\right]$
3. Fixed at outer edge with in-plane force [18.14]  The radial displacement at the outer edge is zero.	Relationship between load and deflection: $\frac{p_1 a^4}{Eh^4} = \frac{5.333}{1-\nu^2}\frac{w_0}{h} + 0.857\left(\frac{w_0}{h}\right)^3$ Maximum radial stress at center: $\sigma_{r0} = \frac{Eh^2}{a^2}\left[\frac{2}{1-\nu^2}\frac{w_0}{h} + 0.5\left(\frac{w_0}{h}\right)^2\right]$ Maximum radial stress at outer boundary: $\sigma_{ra} = \frac{Eh^2}{a^2}\left(\frac{4}{1-\nu^2}\frac{w_0}{h}\right)$